GITAM Deemed to be University

Electrical and Electronics Engineering | Course Code: EE304

Power Electronics & Converters - Comprehensive Syllabus Lecture Notes

Syllabus Focus: EE304 Full Syllabus Comprehensive Lecture Notes Units 1-5

Summary: Complete syllabus lecture notes covering Power Electronics & Converters. Includes deta

I. UNIT-BY-UNIT COMPREHENSIVE SYLLABUS CONTENT

Unit 1: Power Semiconductor Switches

" Syllabus Scope: Thyristors SCR, Triac, Power MOSFET, IGBT, Switching characteristics, Safe Operating Ar

Unit 2: Phase-Controlled Converters

" Syllabus Scope: Single-phase and three-phase half-wave and full-wave controlled rectifiers with R, RL,

EE304: Power Electronics & Converters - Advanced Topics & Assessment

Academic Track: Electrical and Electronics Engineering - Units 3, 4, 5 & Examination Matrix

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II. ADVANCED SYLLABUS UNITS (UNITS 3 TO 5)

Unit 3: DC-DC Switched Mode Regulators

" Buck converter, Boost converter, Buck-Boost topology, Continuous and discontinuous conduction boundary modes

Unit 4: Inverters & Modulation Techniques

" Voltage Source Inverters VSI, Single-phase full-bridge, Three-phase 120 and 180 conduction modes, Sinusoidal

Unit 5: AC Voltage Controllers & Cycloconverters

" Single-phase bidirectional controllers with resistive and inductive loads, Sequence control, Cycloconverter

III. HIGH-YIELD SEMESTER EXAM QUESTION BANK & SOLUTIONS

1. Derive the expression for the average output voltage of a single-phase fully-controlled bridge rectifi
2. Formulate the inductor volt-second balance and capacitor charge balance for a DC-DC Boost regulator in
3. Compare 180-degree and 120-degree conduction modes of a three-phase bridge inverter with line and phas

IV. PRESCRIBED TEXTBOOKS & REPOSITORY CITATIONS

* Ned Mohan, Tore M. Undeland - Power Electronics: Converters, Applications, and Design Wiley 3rd Ed
* Muhammad H. Rashid - Power Electronics: Devices, Circuits, and Applications Pearson 4th Ed

Authorized NoteVault Educational Document " Verified for Registered University Students